

member is an active participant in the crime. So the misconception that using *public* trackers is unsafe is completely unfounded, because using private trackers is just as risky. Perhaps the best place to hide really is in a crowd!

What we're trying to say here is that although you can find anything using the BT network, it's not wise to use it to download pirated content—our government has started tracking down and taking action against the pirates, and you never know who'll be next.

Why Not Stop BT?

Unlike other P2P networks, however, BitTorrent cannot be blamed for the way the technology is being used, so there's very little chance that the protocol will die—it's just a protocol, and there's no one governing body that controls what is on the network (much like there's no one organisation that governs the Internet).

The BitTorrent network is used for a lot of legal downloads as well, and most open source developers release large software files on the BT network to save themselves bandwidth costs in terms of HTTP servers. Linux distros are perfect examples of this. It's not just software; creators of free audio and video projects are also releasing their work over the BT network—so you can legally download movies, music and more for free using BitTorrent, too.

Malicious Content

P2P networks are often associated with malicious content, and this was another reason for BitTorrent's birth. Because you mostly find torrent files on Web sites that host them, it's easy to read feedback provided by previous users who downloaded the file and gauge whether the file is indeed what it claims to be. Sometimes, due to errors during file creation, you might receive a bad file. This writer once got a corrupted Ubuntu CD ISO file from the BT network, all because some peer had a corrupted segment of the file. These occurrences are rare, though, and can even be prevented (see box *PeerGuardian*). This writer also remembers downloading what was supposedly "freeware" from earlier networks such as KaZaA, and getting viruses instead. In fact, people using earlier networks would cross their fingers when downloading something, hoping that the file was what it claimed to be—with instances of people trying to download *My Fair Lady* and ending up with pornography, and so on.

Thankfully, when using the BT network, you can just go back to the site you downloaded the file from and leave a comment to prevent others from being duped. In fact, you will almost always find comments from users for almost every torrent downloaded. Those who download pirated content find it easy to gauge the authenticity of the download by looking at the number of seeds and peers for that file—the larger the swarm, the higher the probability that the content is what it claims to be.

Clients And Sites

In order to use the BitTorrent network, you need to have a BT client, and you can get the



PeerGuardian

PeerGuardian and PeerGuardian 2 are applications made by Phoenix Labs (www.phoenixlabs.org) that can stop your computer from connecting to banned IPs. Usually, IPs that are known to be sending out malicious data, junk data, etc., are logged to an IP Blocklist, and this is updated on a server. When you have PeerGuardian running on your computer, it updates itself with this Blocklist and stops you, or any applications running on your computer, from connecting to this list of banned IPs.

When it comes to the BitTorrent network, PeerGuardian helps by filtering out peers who are uploading junk data for popular files. Unfortunately, most popular files are pirated content, and the rumours floating about on the Net suggest that organisations such as the MPAA and RIAA are using the technique of sending out junk data disguised as legitimate parts of popular torrents to corrupt the files that downloading peers get, or flood the network so that it takes much longer for the file to completely download. This happens because torrent clients do an immediate hash check for each part of the file they download. If the file chunk is filled with bad data, the part is discarded and re-downloaded from another source. This is why a lot of people have started complaining about receiving a lot of junk data and taking days to complete even small downloads.

By blocking such IPs, PeerGuardian seems to be helping piracy. However, it also allows you to prevent your computer from connecting to adware-serving hosts, spyware servers, US government IPs (so they don't set out to get you, in case they're looking) and even educational institutions (where a lot of script kiddies sit and use college resources to try and hack PCs across the globe). This software will also help you avoid junk data on the torrent network when downloading legal torrents. We're forced to accept the bad with the good and recommend it. Remember that the blocks are computer-wide, and not just limited to your torrent client, but all protocols and ports.

original BitTorrent client from www.bittorrent.com. However, if you want more advanced torrent clients, there are: BitComet (www.bitcomet.com), Azureus (<http://azureus.sourceforge.net>), and µTorrent (www.utorrent.com).

In order to download torrent files and legal content, you need to check out the BitTorrent home page at www.bittorrent.com and Legal Torrents at www.legaltorrents.com. Most other places on the Web will have both legal and illegal downloads listed, and you might find it hard to distinguish between the two. They're not hard to find, though, and Google searches for torrents will almost always point you to such sites. Remember, you've been warned, so be extra careful when downloading content from any P2P network, or you could unwittingly break the law. You should also be careful if you have youngsters in your household, because a lot of torrent sites are supported by advertising from the pornographic industry, and are obviously not fit for use on family computers. For the same reason, you should resist the temptation to access these sites from your office as well—you could get your company in trouble too.

If you have broadband, you're going to love the speeds of the downloads that the BT network offers. (Those of you on limited data transfer plans should remember that *both* upload and download data counts in the final tally). Enjoy your BT experience! ☑

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